(A) Categorical Derivation

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FROM CATEGORICAL ENTROPY
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S = k_B M ln n  with M = 3N

States: n = V/V_0

S = 3N k_B ln(V/V_0)

From dS/dV = P/T:

3N k_B / V = P/T

=> PV = 3N k_B T

(Factor 3 = spatial dimensions)
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(B) Oscillatory Derivation

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FROM EQUIPARTITION
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Each particle: <(1/2)mv_x^2> = (1/2)k_B T

Total kinetic energy:
U = (3/2) N k_B T

Pressure from momentum flux:
P = 2U / (3V)
P = (2/3V) * (3 N k_B T / 2)

P = N k_B T / V

=> PV = N k_B T
```

(C) Partition Derivation

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FROM PARTITION FUNCTION
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Cells: n = V / lambda^3
(lambda = thermal de Broglie wavelength)

Z = (1/N!) * (V/lambda^3)^N

Free energy: F = -k_B T ln Z

P = -dF/dV at constant T, N

P = k_B T * (N / V)

=> PV = N k_B T
```
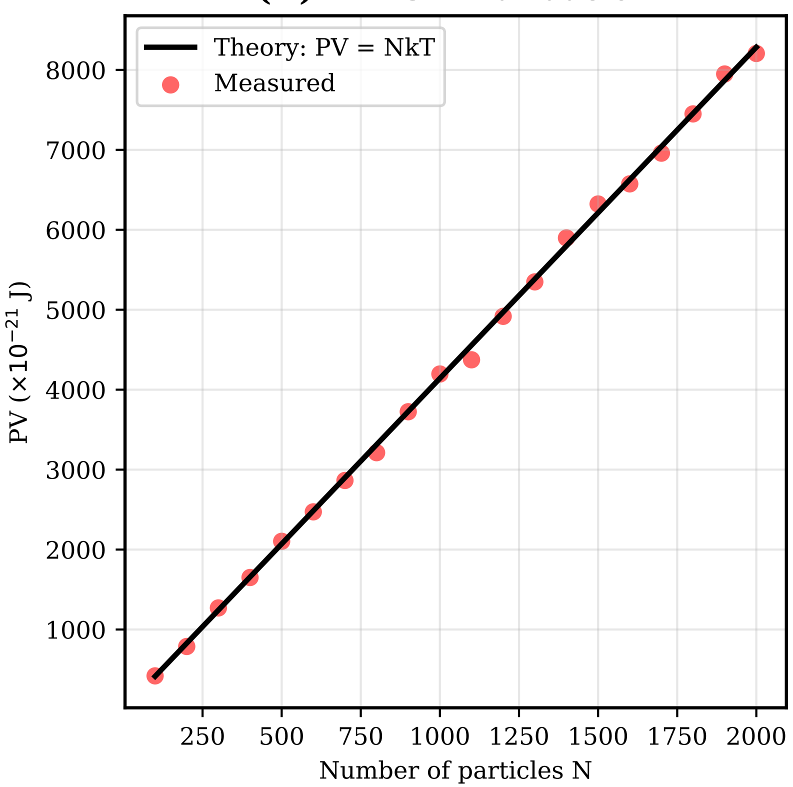
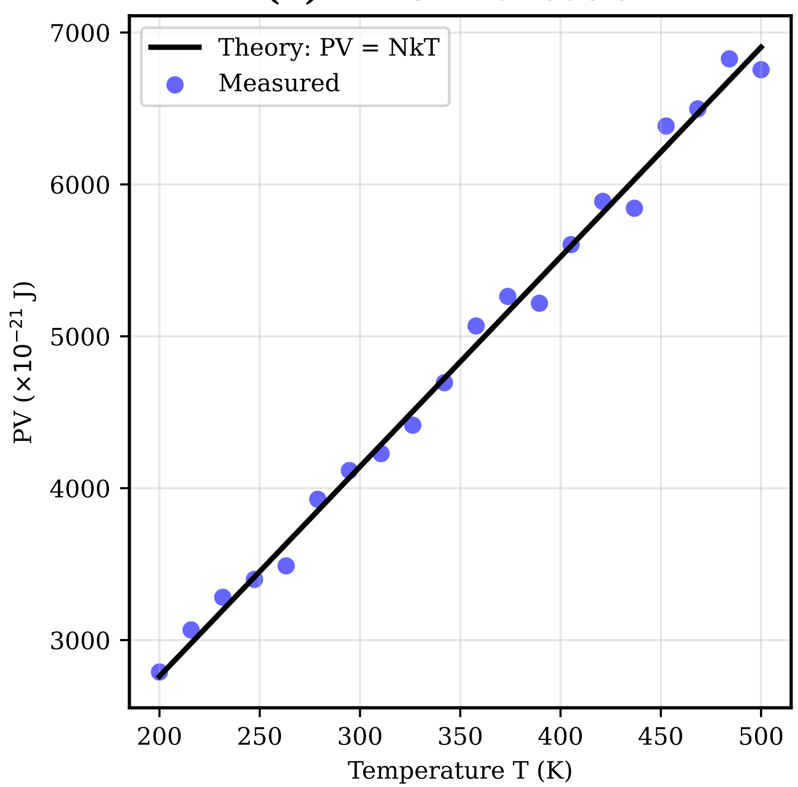

(D) PV vs N Validation



(E) PV vs T Validation



(F) Ideal Gas Law Summary

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IDEAL GAS LAW VALIDATION
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Three Independent Derivations:

1. CATEGORICAL:
   S = k_B M ln n  -->  PV = NkT
   (Entropy counting)

2. OSCILLATORY:
   Equipartition   -->  PV = NkT
   (Energy averaging)

3. PARTITION:
   Z = (V/lambda^3)^N  -->  PV = NkT
   (Statistical mechanics)

ALL THREE DERIVATIONS AGREE:
   PV = NkT

VALIDATION: PASS

The ideal gas law is the categorical
balance condition for bounded systems.
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